

Paper Presentation

Scene Coordinate Reconstruction:
Posing of Image Collections via Incremental Learning of a Relocalizer

B10901151 林祐群

Instructor: Homer H. Chen

Contents

- [1] Brachmann, E., Wynn, J., Chen, S., Cavallari, T., Monzpart, '., Turmukhambetov, D., & Prisacariu, V. (2024). Scene Coordinate Reconstruction: Posing of Image Collections via Incremental Learning of a Relocalizer. *European Conference on Computer Vision*.
- [2] Brachmann, E., Cavallari, T., & Prisacariu, V. (2023). Accelerated Coordinate Encoding: Learning to Relocalize in Minutes using RGB and Poses. In *CVPR*.
- [3] Saeed, S. U., Fu, Y., Stavrinides, V., Baum, Z. M. C., Yang, Q., Rusu, M., Fan, R. E., Sonn, G. A., Noble, J. A., Barratt, D. C., & Hu, Y. (2022). Image quality assessment for machine learning tasks using meta-reinforcement learning. *Medical image analysis*, 78, 102427. <https://doi.org/10.1016/j.media.2022.102427>

Outline

- A. Overview - Goals & Merits
- B. Background Comparison - Visual Relocalization
- C. ACE
- D. Background - Reconstruction
- E. ACE0 (ACE Zero) Framework
- F. Evaluation Metrics
- G. Experiments & Results
- H. Appendix - IQA with meta-reinforcement learning
- I. References

Overview

- **Task to address:**

Estimating camera parameters from a set of images depicting a scene

- **Current **feature**-based SfM with incremental reconstruction:**

Repeat triangulation of sparse 3D points and registration of more camera views to the sparse point cloud.

- **Main idea:**

Re-interpret incremental SfM as an iterated **application and refinement of a visual relocalizer**, that is, of a method that registers new views to the current state of the reconstruction.

Find alternatives (from their previous research) for visual localizers that are not rooted in local feature matching.

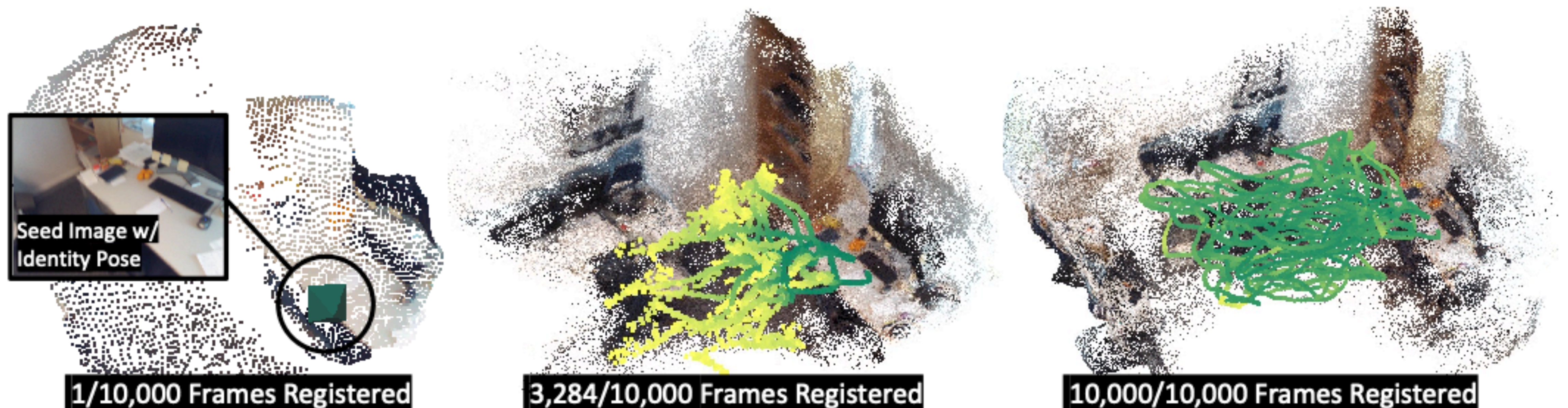
- **Results:**

Can be applied to multiple thousand unsorted images, without pose priors, and reconstructs them within a few hours

Overview

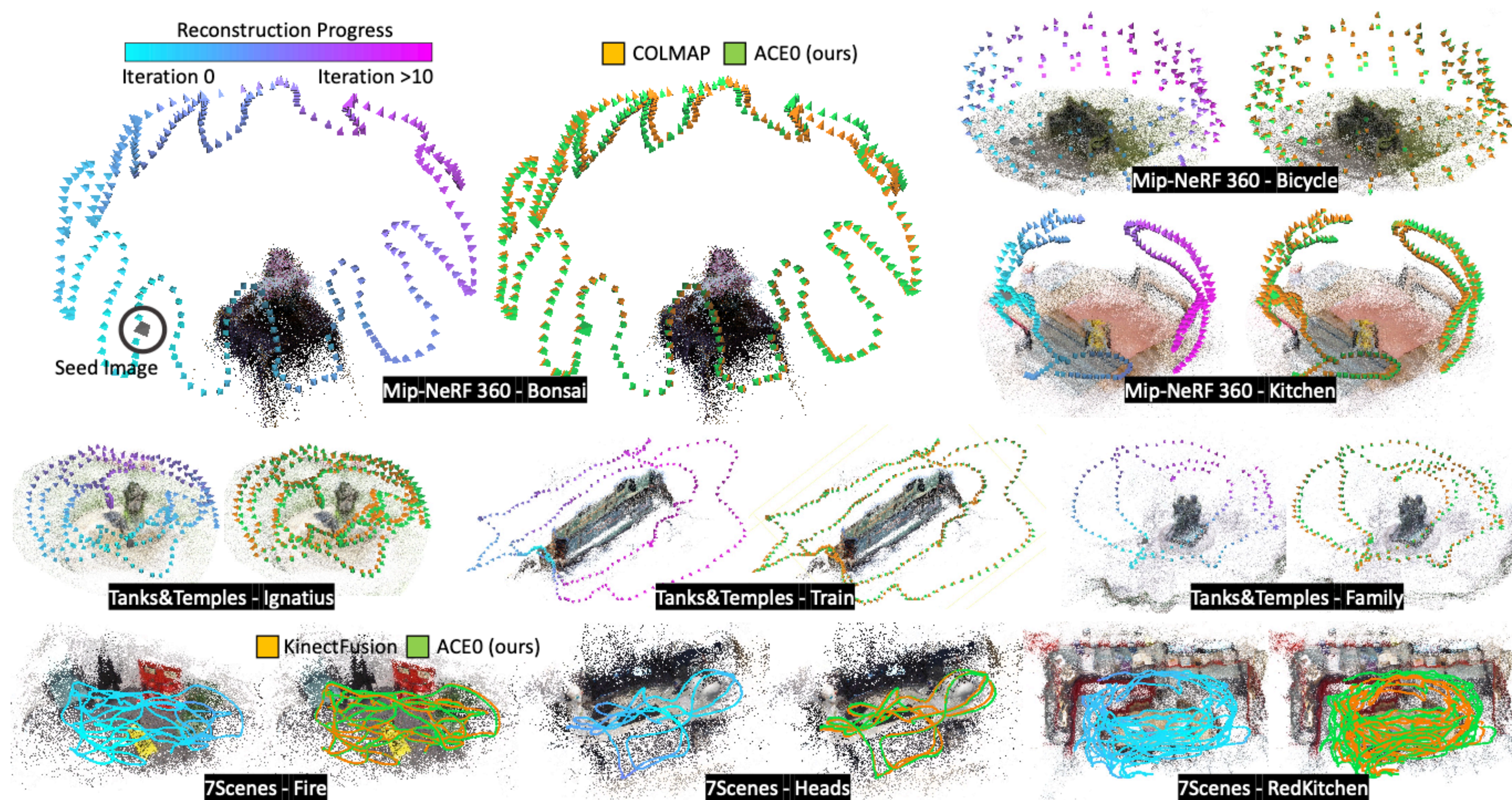
Starting from a single image and the identity pose, the learning-based visual relocalizer allows to estimate the poses of more views, and the additional views allow refinement of the relocalizer.

- Camera positions are color coded by relocalization confidence from yellow (low) to green (high)



3 out of 6 iterations, 10k images have been posed in roughly 1 hour on single GPU

Overview



Background - Visual Relocalization

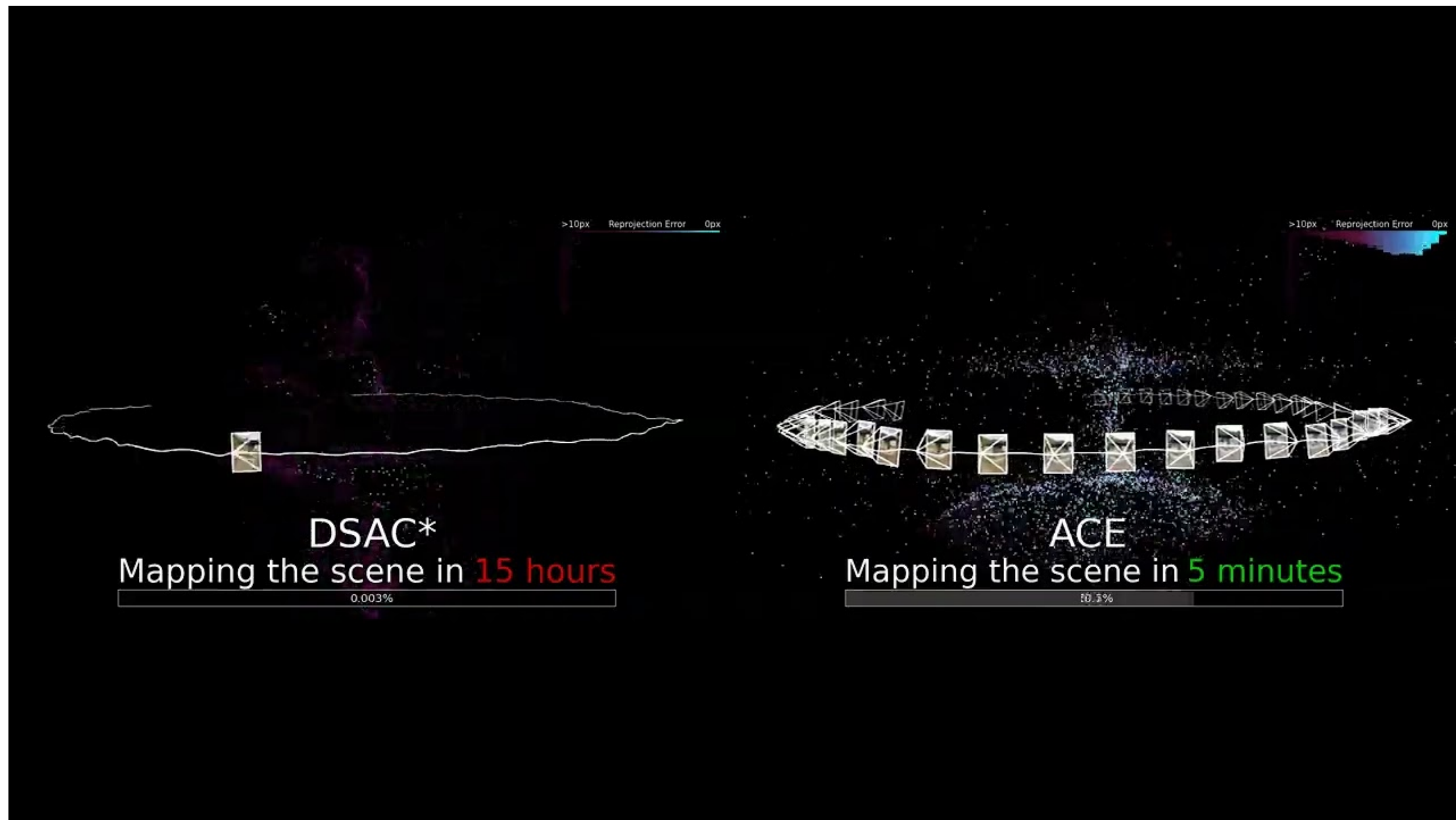
Feature-Matching Approaches

- Builds an explicit 3D reconstruction of a scene using SfM software
- Optimizes over image-to-image matches
- Representing the scene as a 3D point cloud with high dimensional descriptors

Scene Coordinate Regression

- Learns an implicit representation of the scene via gradient descent.
=> Learning-based
=> Originally slow
- Regresses image-to-scene correspondences.
- Encode the scene into a lightweight neural network (4MB for ACE)

Visual Relocalizer - ACE



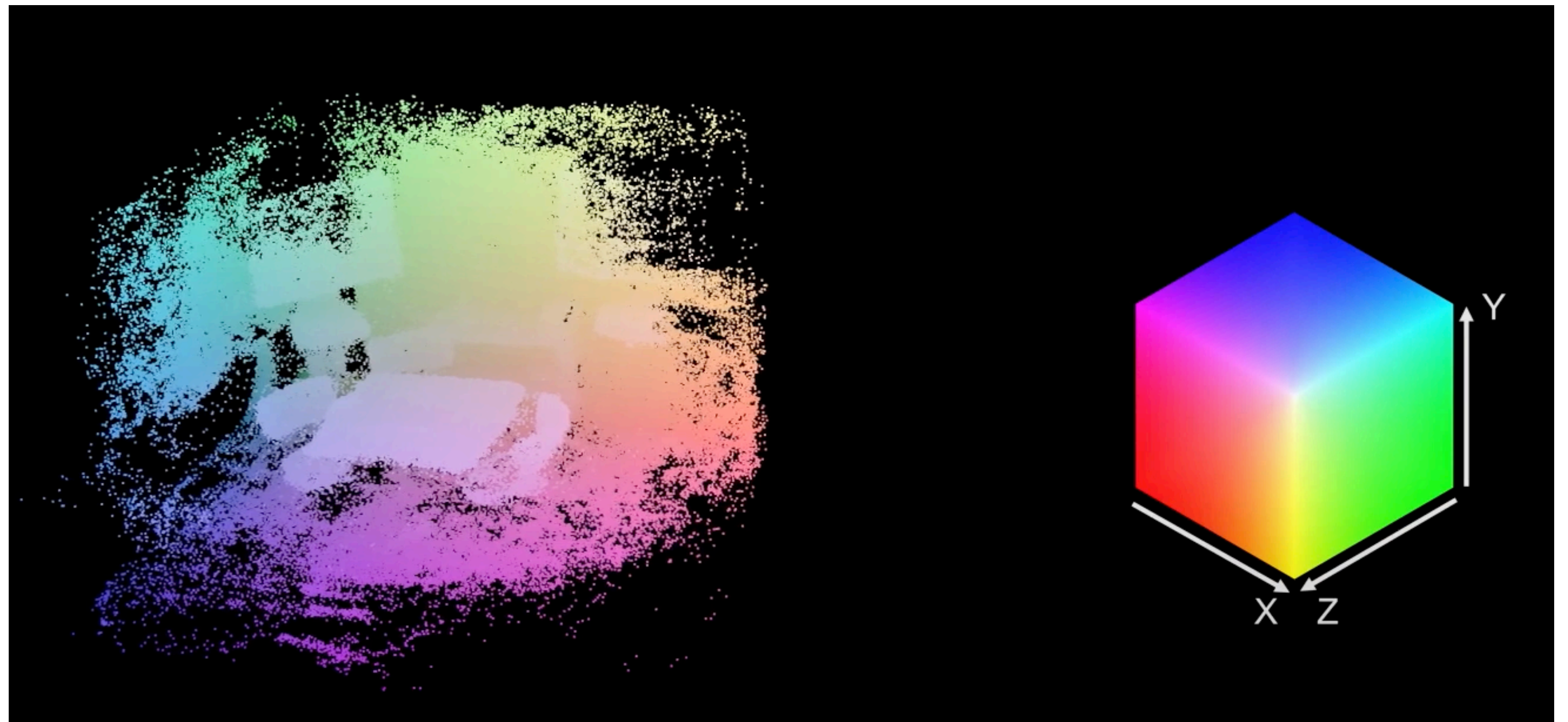
Time is really the only capital that any human being has, and the only thing he can't afford to lose.

—Thomas Edison

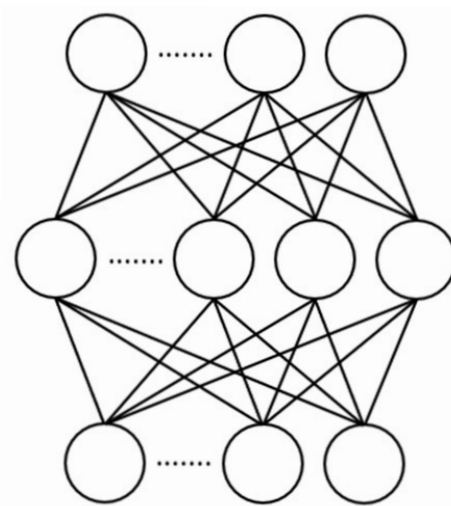
Scene Coordinate Regression - 1

Each point has a unique 3D coordinate in the scene coordinates.

=> Visualize them by mapping the XYZ coordinate of each point to the RGB cube



Scene Coordinate Regression - 2

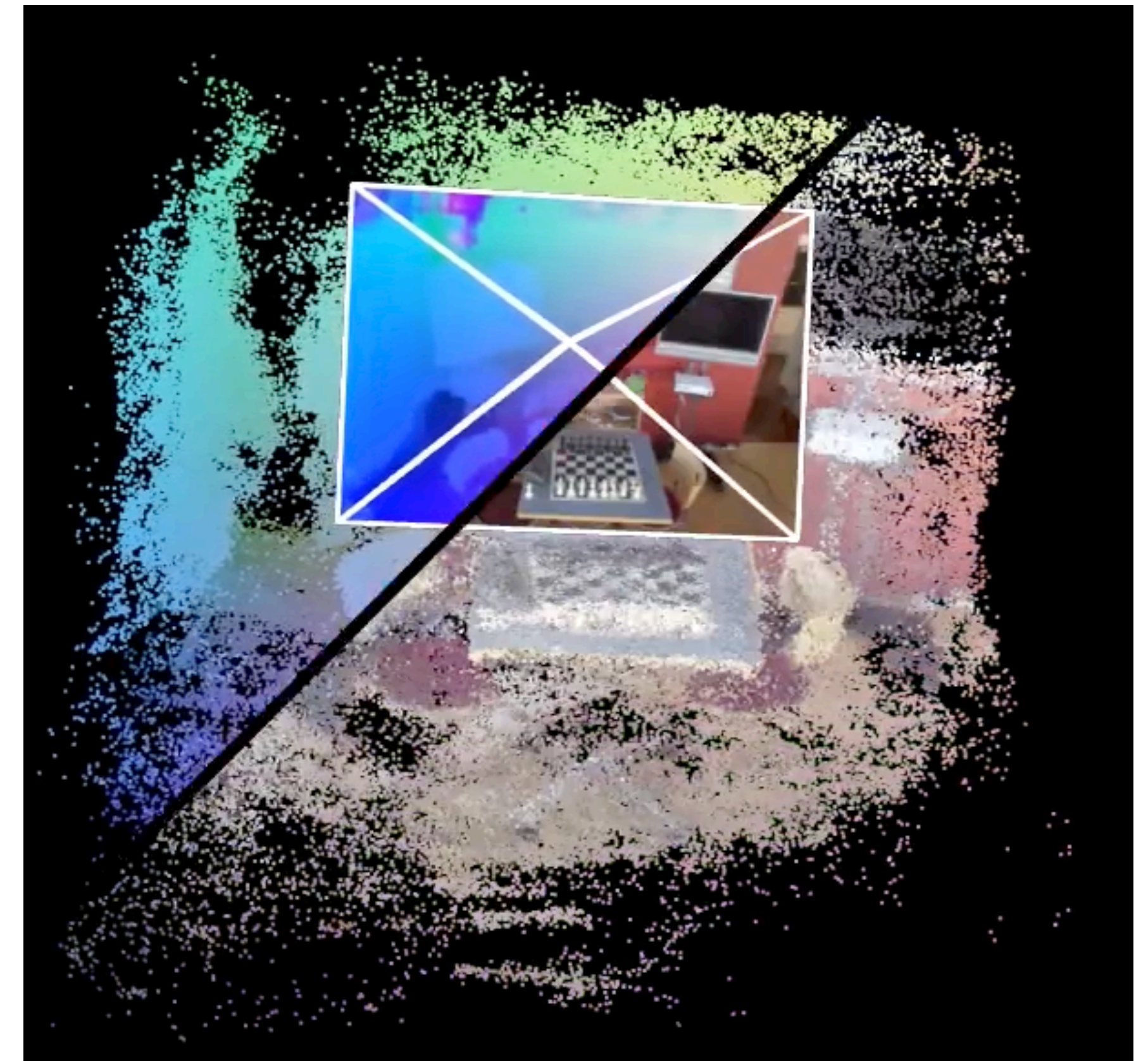
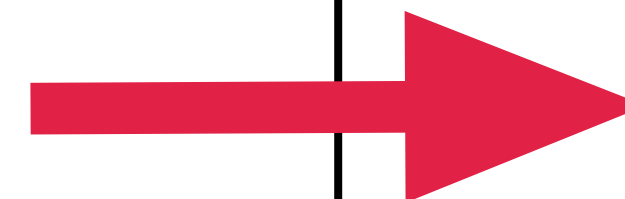


Query Image

2D

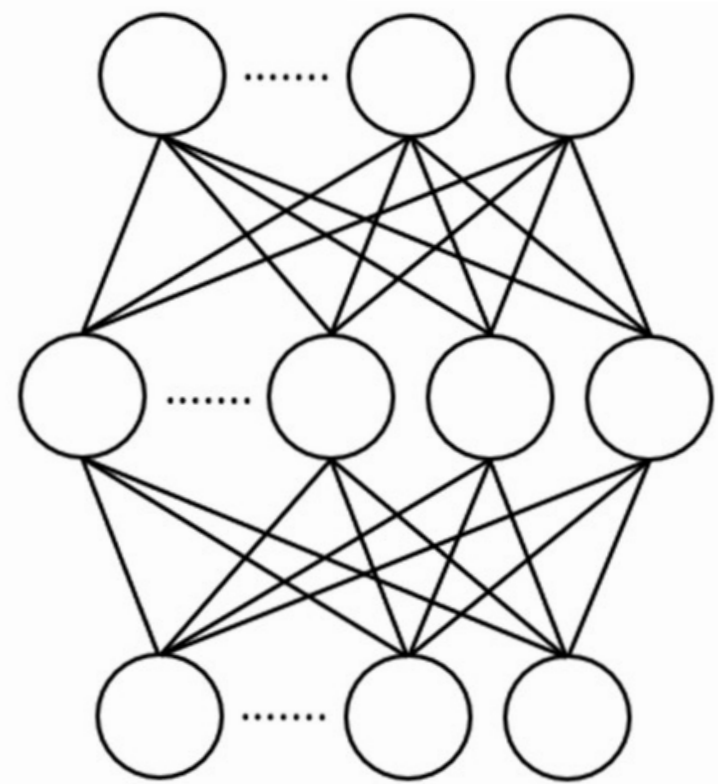
3D

Scene Coordinate
Prediction (of each
pixel in the image)



Inference:
RANSAC + PnP

Comparison: DSAC*



Train with
reprojection error

p_i : 2D Pixel Position
 π : Projection Function
 y_i : 3D Scene Coordinate
 h^* : GT Mapping Pose

Reprojection Error: $l_i = \| p_i - \pi(y_i, h^*) \|_1$

=> single view constraints

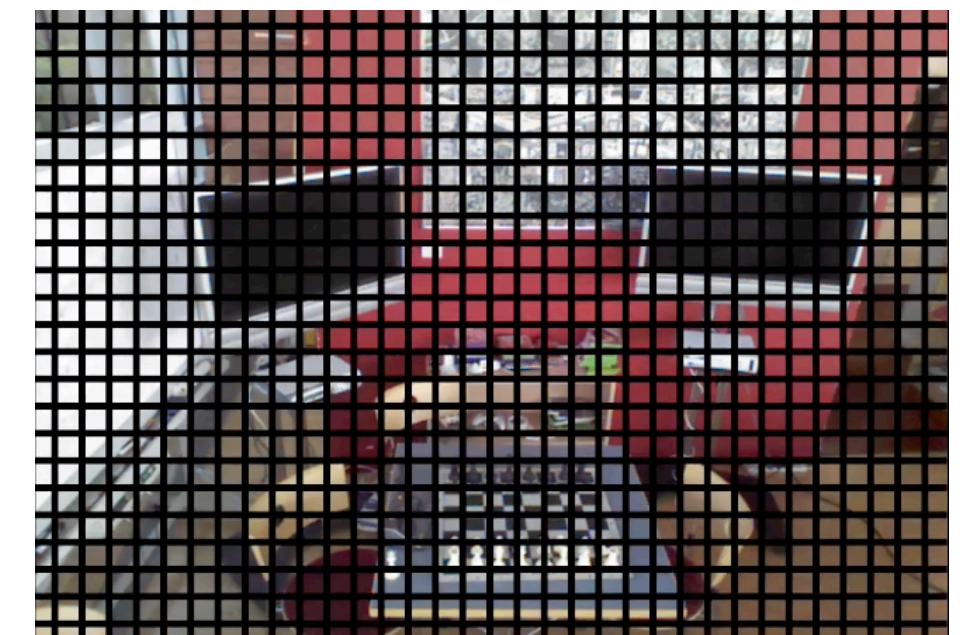
=> depth is ill-defined ?



Through training
with different images

Converge to a scene representation that
is consistent across views, with scene
coordinates having their correct depth

Optimize one mapping
image at a time

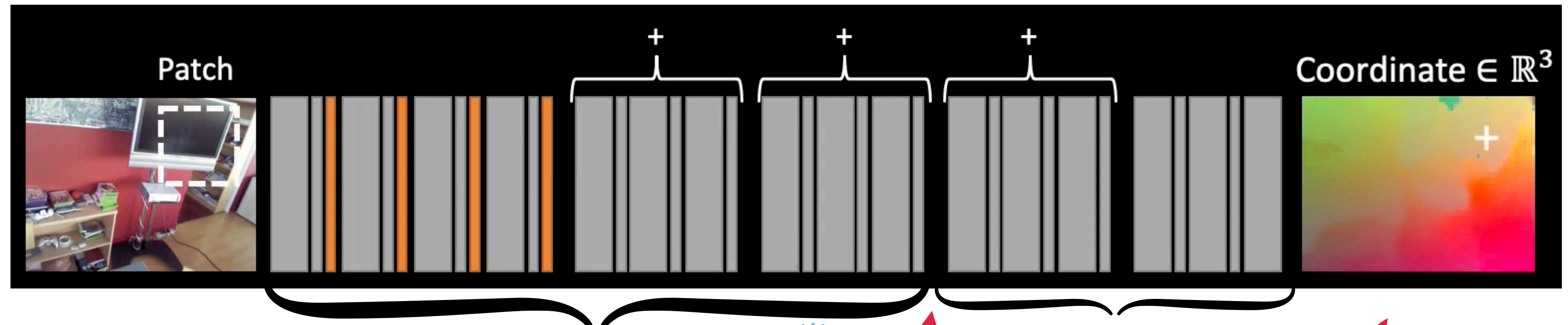


$$l_i \approx l_{i+1}$$

Coarse gradient
=> need small learning rate

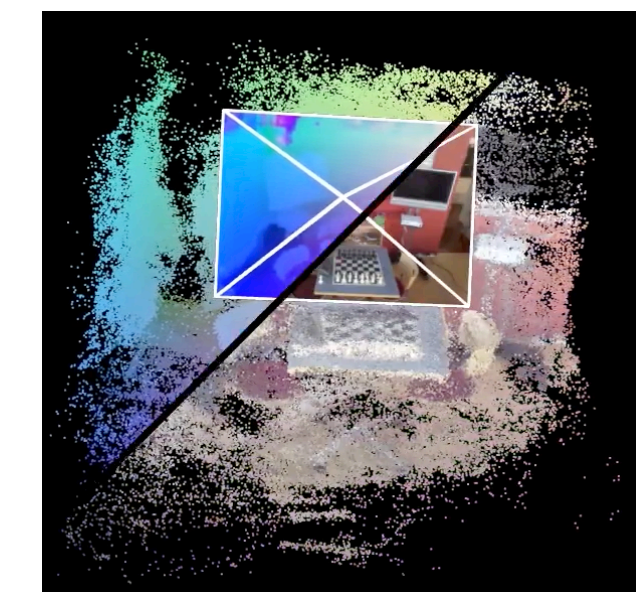
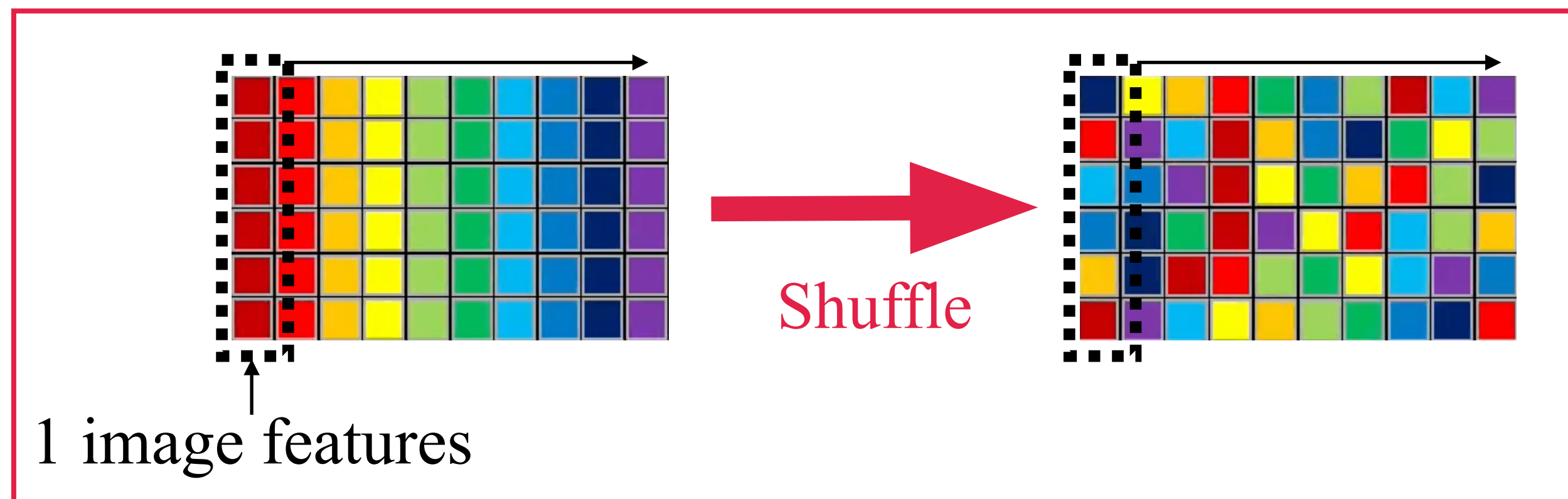
Comparison: ACE

Fully convolutional, with limited receptive field



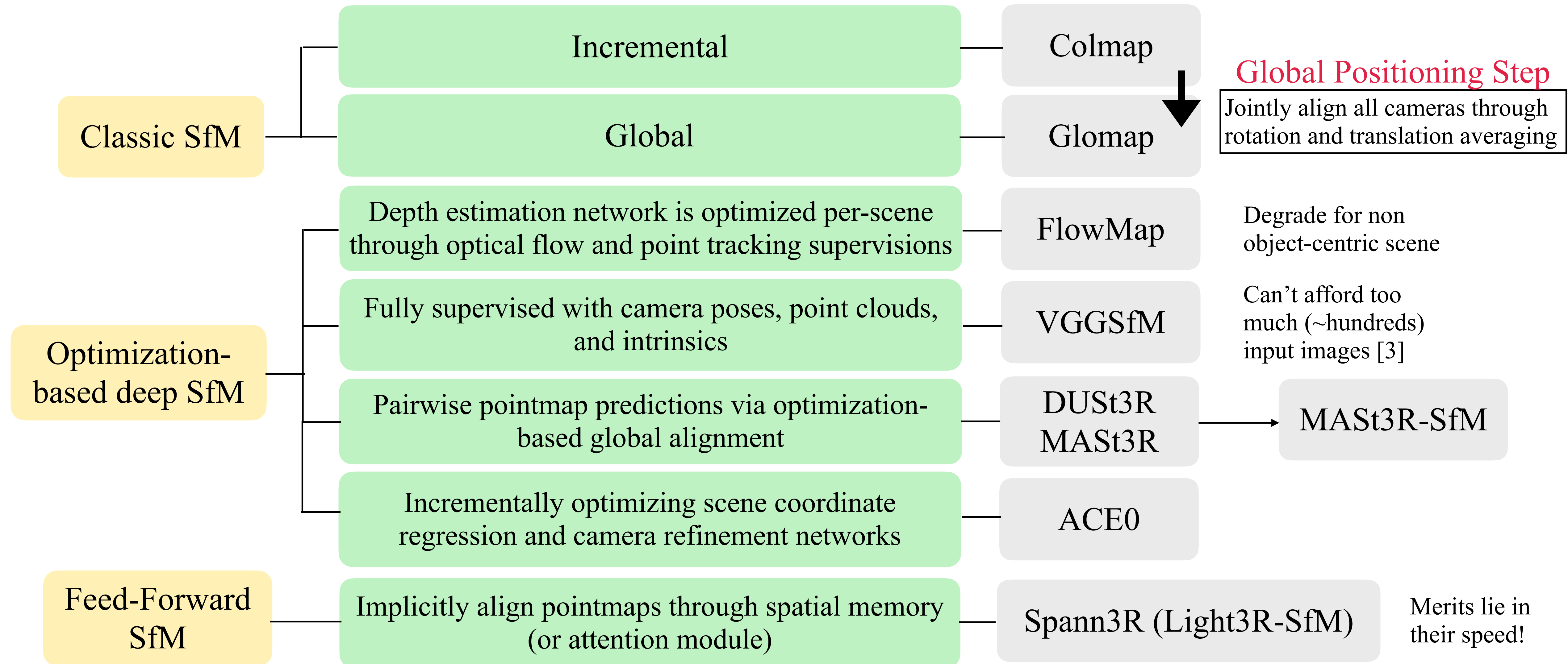
CNN Feature Backbone ❄️
(Pre-trained Scene-agnostic)

MLP Regression Head 🔥



Inference: RANSAC + PnP

Background - Reconstruction



ACE0 Framework

Assumption: **No** particular image order or any prior knowledge about the pose distribution

Input: A set of RGB images, $I = \{I_i\}$ where i is image index

To estimate:

1. A set of camera parameters, $H = \{(K_i, T_i)\}$
2. The 3D structure of the scene

$\Rightarrow \mathbf{p}_{ij} = \pi(K_i, T_i, \mathbf{y}_{ij})$, with projection function π

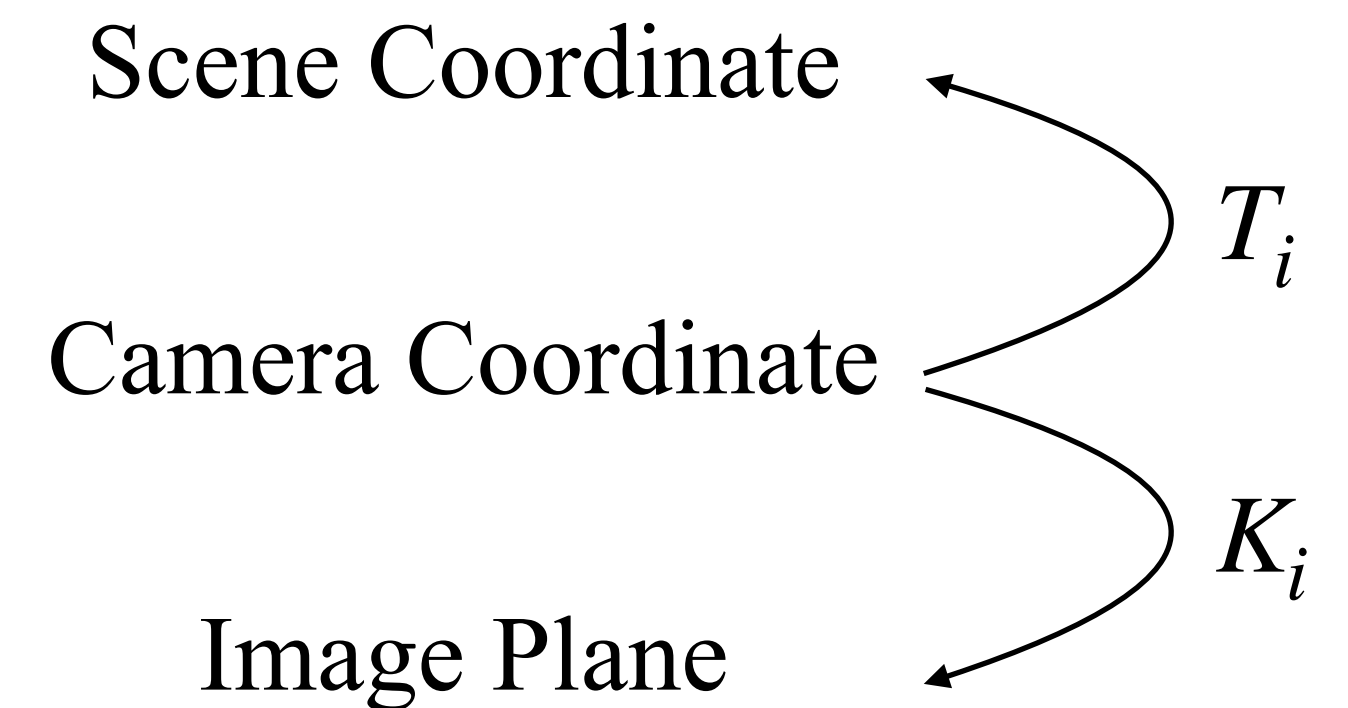
Scene Representation: A learnable f_{SCR} s.t. $\mathbf{y}_{ij} = f_{SCR}(\mathbf{p}_{ij}, I_i)$

$\Rightarrow$ maps an image patch of image I , centered around pixel position $\mathbf{p}_{ij}$ to a scene coordinate

Recover Image's Camera Pose T_i : using a pose solver g

$\Rightarrow T_i = g(K_i, \{\mathbf{p}_{ij}, \mathbf{y}_{ij}\})$, further combines with PnP solver & RANSAC

- T_i : rotation, translation (3x4 matrix)
- K_i : calibration parameters (3x3 matrix)
- Each pixel j in image i with 2D pixel position $\mathbf{p}_{ij}$, and its corresponding coordinate in 3D $\mathbf{y}_{ij}$.



ACE0 Framework

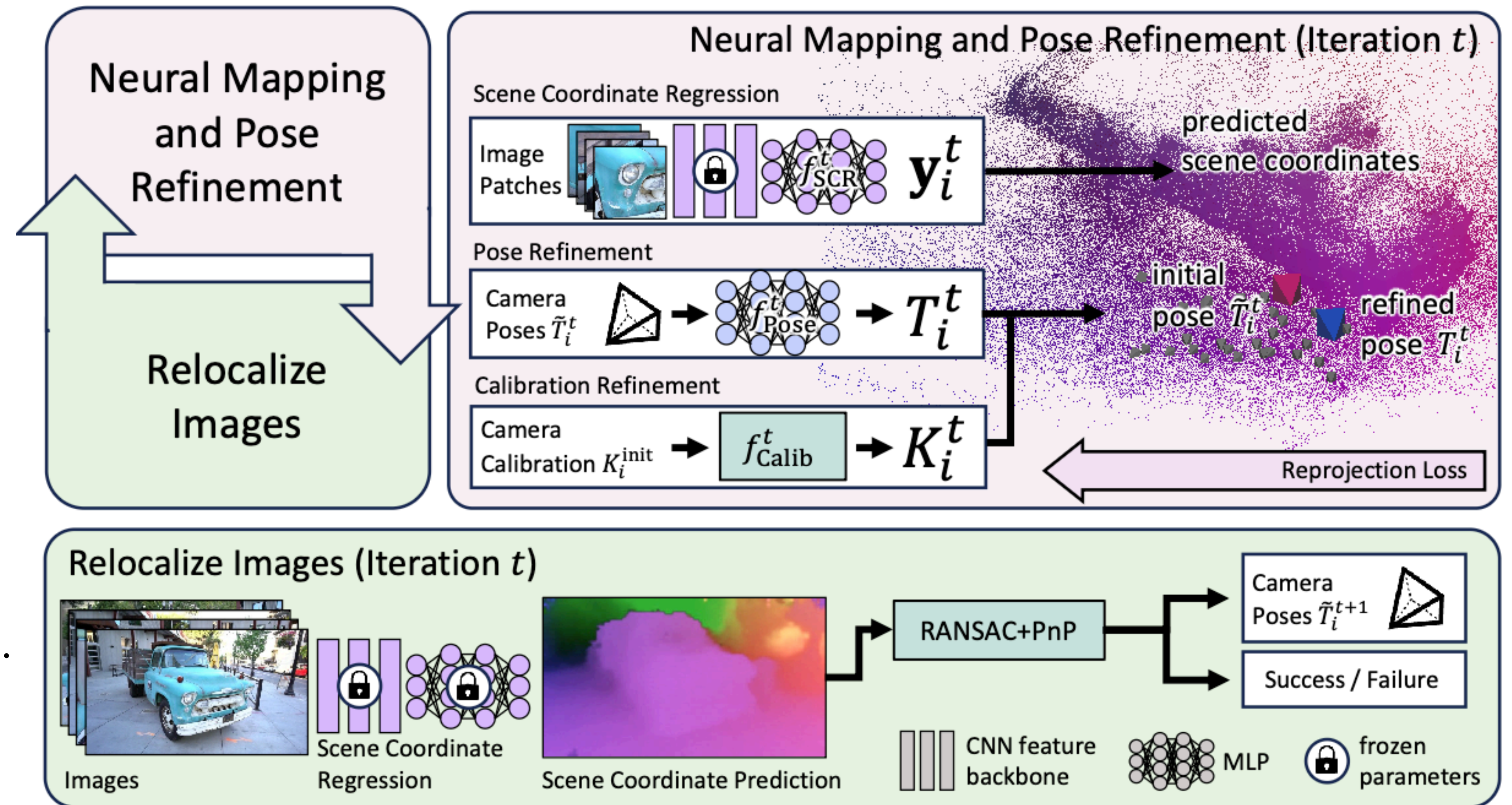
Normally, Scene Coordinate Regression models are trained in supervised fashion. (Need GT)
=> ACE0 trains them using **self-supervise** learning.

Loop between:

1. Learning a reconstruction from the current set of images and poses (“**neural mapping**”)

&

2. Estimating poses of more images (“**relocalization**”).



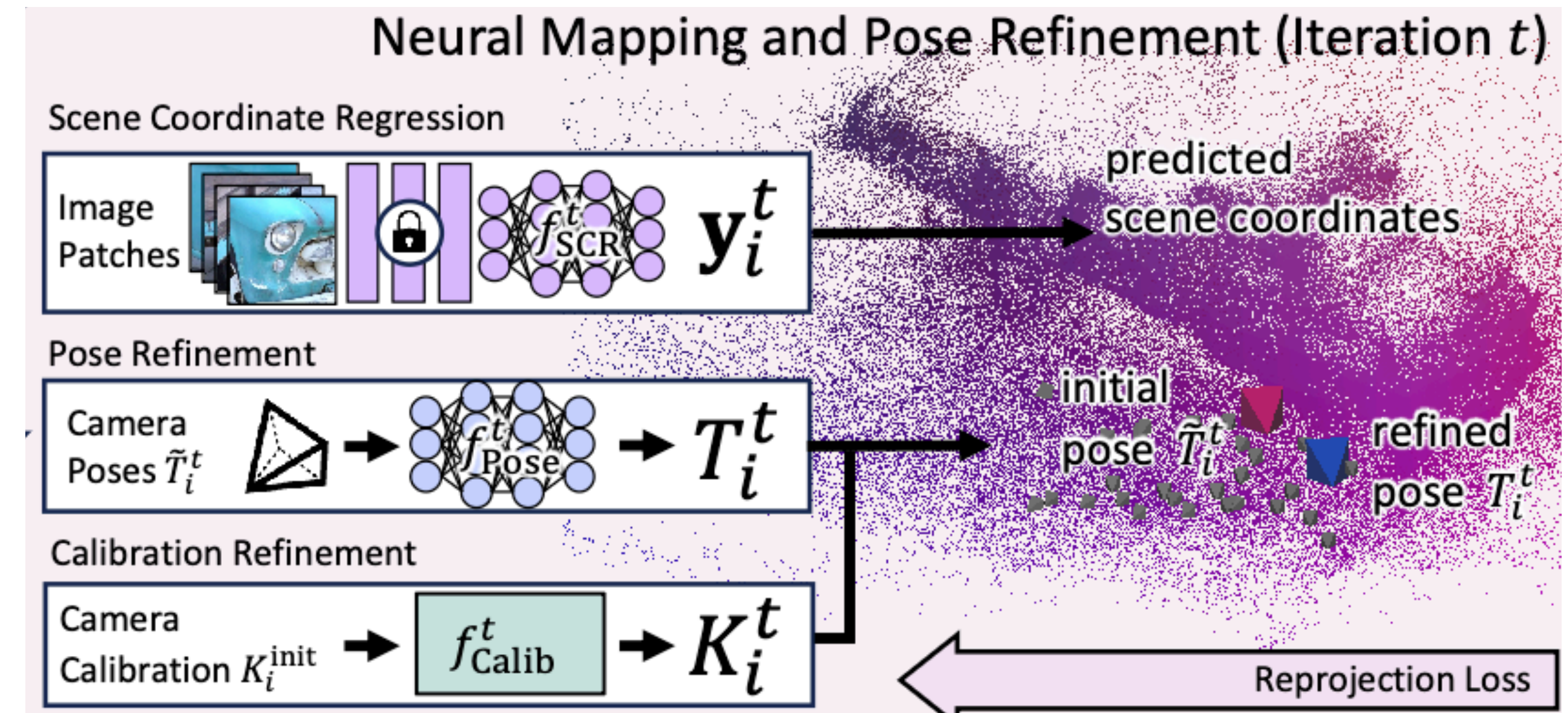
ACE0 Framework - Neural Mapping

- For **register images** $I_{Reg}^t \subset I$, where pseudo GT (camera parameters K_i^t , T_i^t) are already estimated.

=> train the scene model by minimizing pixel-wise Reprojection error:

$$\begin{aligned} & \sum_{I_i \in I_{Reg}^t} \sum_{j \in I_i} \|\mathbf{p}_{ij} - \pi(K_i^t, T_i^t, \mathbf{y}_{ij}^t)\| \\ &= \sum_{I_i \in I_{Reg}^t} \sum_{j \in I_i} \|\mathbf{p}_{ij} - \pi(K_i^t, T_i^t, f_{SCR}^t(\mathbf{p}_{ij}, I_i))\| \end{aligned}$$

=> Optimize using stochastic gradient descent, and ACE



ACE0 Framework - Refinement

- GT poses T_i^{GT} are unknown during training
 $\Rightarrow$ utilize T_i^t , estimates based on earlier iterations of the reconstruction (**inaccurate** though)

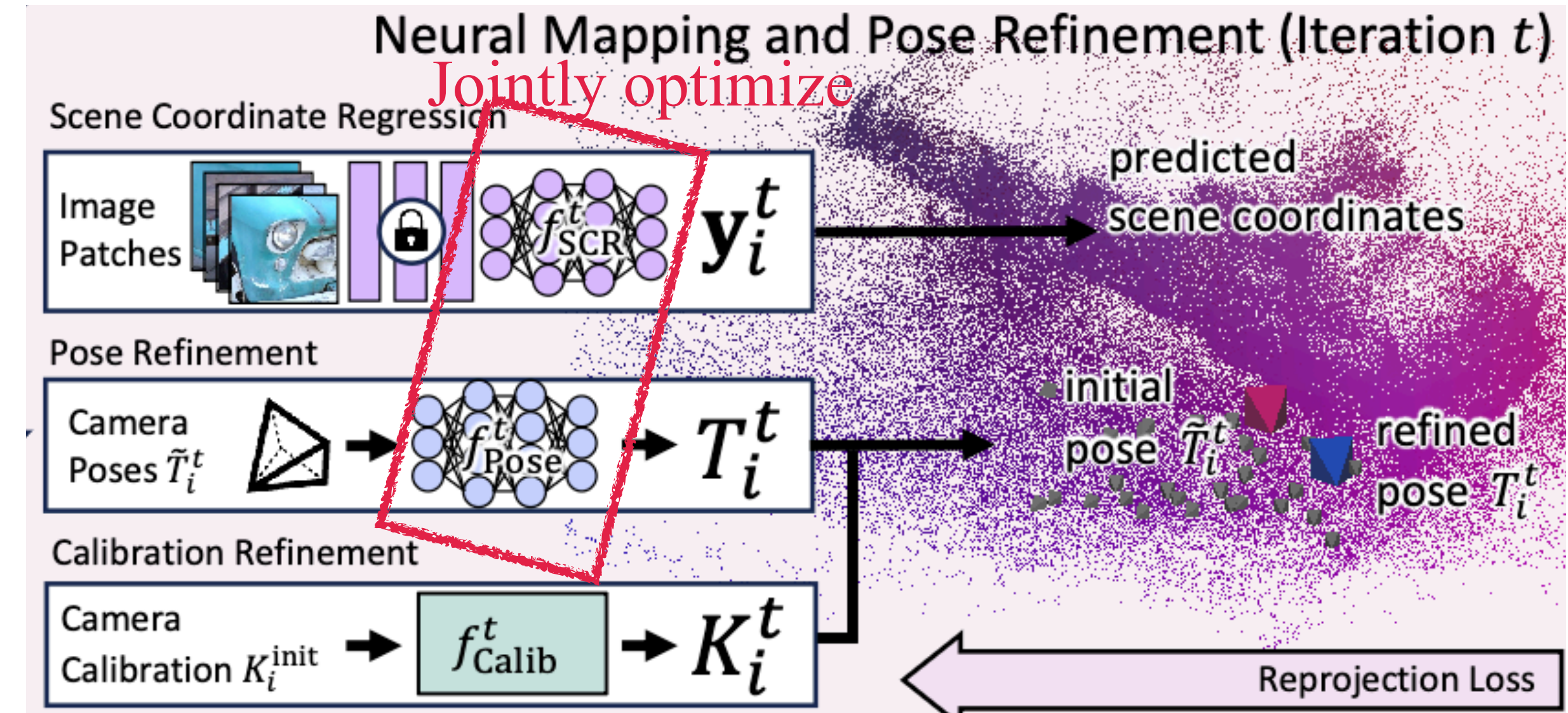
$\Rightarrow$ using MLP for refinement $T_i^t = f_{Pose}^t(\tilde{T}_i^t)$



- Refine calibration from heuristics $K_i^t = f_{Calib}^t(K_i^{init})$

f_{Calib}^t entails a single learnable parameter α_t s.t. focal length $f_i^t = f^{init} \cdot (1 + \alpha^t)$

α estimation is carried over across iterations

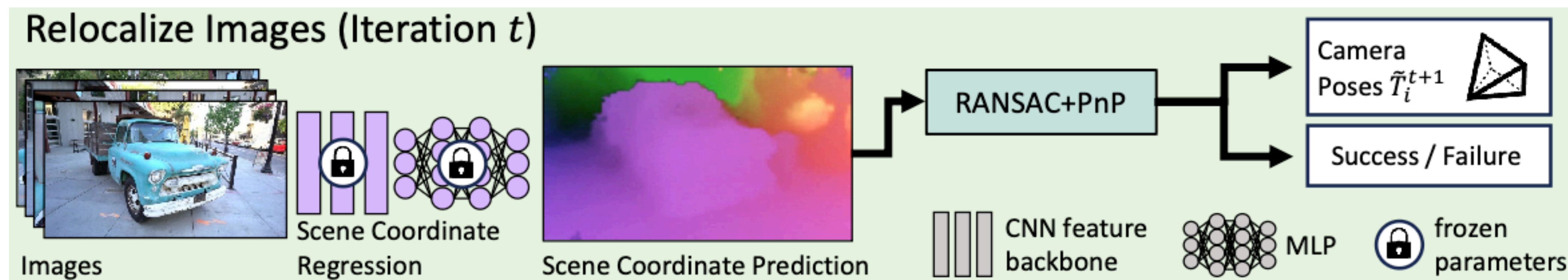


ACE0 Framework - Relocalization

Pass **all images** in I to the scene coordinate regressor to gather 2D-3D correspondences.

=> solve their poses using RANSAC and PnP

$$\tilde{T}_i^{t+1}, s_i^{t+1} = g(K_i^t, \{(\mathbf{p}_{ij}, \mathbf{y}_{ij}^t)\}), \text{ where } s_i^{t+1} \text{ is a confidence score.}$$



ACE0 Framework - Initialization

Start the reconstruction with **one** image, set the seed pose T_{seed}^0 to identity.

$\therefore$ Reprojection error is ill-conditioned without multiple views constraining the depth.

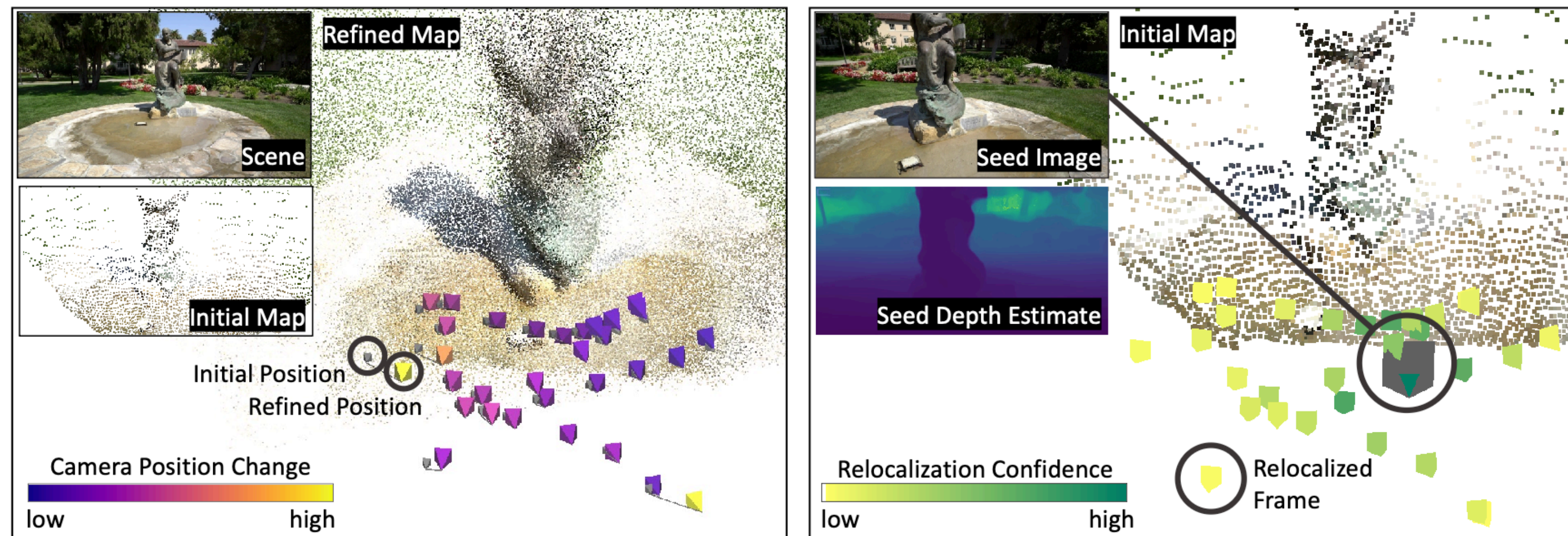
$\therefore$ It's needed to optimize a different objective in the seed iteration.

=> Train the initial scene coordinate regression network f_{SCR}^0 by optimizing: $\sum_{j \in I^0} \|\hat{\mathbf{y}}_{ij} - \mathbf{y}_{ij}\|$

Where $\hat{\mathbf{y}}_{ij} = d_{ij}(K_{seed}^{init})^{-1}\mathbf{p}_{ij}$

d_{ij} is a depth value predicted for pixel j of image i (using ZoeDepth).

Depth estimation is only used in this step.



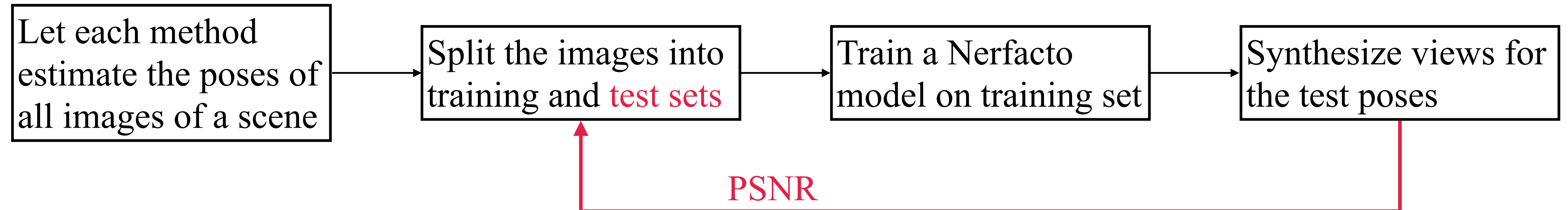
Evaluation Metrics

Evaluate on 3 datasets:

1. A relocalization dataset - 7-Scenes
2. A view synthesis dataset - Mip-NeRF 360
3. A construction dataset - Tanks and Temples

$\therefore$ Pseudo ground truth is estimated rather than measured $\therefore$ Comparing poses are problematic.

$\Rightarrow$ Gauge the pose quality in a self-supervised way, using novel view synthesis.



$\Rightarrow$ Can be applied to arbitrary datasets. ($\therefore$ It doesn't need GT geometry)

Experiments & Results - 7-Scenes

Start from KinectFusion poses

	Frames	Pseudo Ground Truth		All Frames				200 Frames			50 Frames	
		Kinect Fusion	COLMAP (default)	COLMAP (fast)	DROID-SLAM [†] [90]	ACE0 (ours)	KF+ACE0 (ours)	BARF [54]	NoPE-NeRF [†] [10]	ACE0 (ours)	DUST3R [97]	ACE0 (Ours)
Chess	6k	19.6	23.6	23.5	19.3	23.3	23.0	12.8	12.6	22.7	18.9	19.2
Fire	4k	19.2	22.6	22.6	13.0	22.3	22.3	12.7	11.8	22.1	18.8	19.5
Heads	2k	17.0	18.8	18.9	17.6	18.8	19.1	10.7	11.8	19.9	18.4	21.3
Office	10k	18.9	21.4	21.6	failed	21.1	21.5	11.9	10.9	19.8	12.5	13.7
Pumpkin	6k	19.9	24.1	23.8	18.3	24.1	23.8	19.6	14.2	24.7	21.7	22.3
RedKitchen	12k	17.6	21.4	21.4	10.9	20.8	20.9	11.6	11.2	18.9	13.8	13.7
Stairs	3k	19.0	16.7	21.0	13.0	17.7	19.9	15.8	15.9	18.8	15.3	15.4
Average		18.7	21.2	21.8	N/A	21.2	21.5	13.6	12.6	21.0	17.1	17.9
Avg. Time		realtime	38h	13h	18min	1h	7min	8.5h	47h	27min	4min*	16min

[†]Method needs sequential inputs. *Results on more powerful hardware > 0.5 dB better within ± 0.5 dB > 0.5 dB worse > 1 dB worse

Experiments & Results - Tanks and Temples

		Frames	DROID-SLAM [†] [90]				Frames	DROID-SLAM [†] [90]				Sparse COLMAP + Reloc + BA	
			COLMAP (default)	Reality Capture	SLAM [†] [90]	ACE0 (ours)		COLMAP (fast)	Reality Capture	SLAM [†] [90]	ACE0 (ours)	COLMAP + Reloc + BA	Sparse COLMAP + ACE0 (ours)
Training	Barn	410	24.0	21.2	19.0	16.5	12.2k	24.4	16.9	13.5	17.7	26.3	25.1
	Caterpillar	383	17.1	15.9	16.6	16.9	11.4k	18.6	17.9	18.9	18.6	18.7	18.8
	Church	507	18.3	17.6	14.3	17.2	19.3k	12.1	-	11.5	16.5	18.5	17.3
	Ignatius	264	20.1	17.7	17.8	19.8	7.8k	20.8	18.6	19.1	20.7	20.9	20.7
	Meetingroom	371	18.6	18.1	15.6	18.0	11.1k	19.4	18.2	17.1	16.6	20.8	20.3
	Truck	251	21.1	19.0	18.3	20.1	7.5k	23.6	19.1	20.6	23.0	23.4	23.1
	Average	364	19.9	18.2	16.9	18.1	14.6k	19.8	18.2	16.8	18.9	21.4	20.9
Avg Time			1h	3min	5min	1.1h		74h	14h	18min	2.2h	8h	1.8h
Intermediate	Family	152	19.5	18.8	17.6	19.0	4.4k	21.2	19.8	19.8	18.0	21.3	21.3
	Francis	302	21.6	20.7	20.7	20.1	7.8k	19.9	20.4	21.8	21.7	22.5	22.7
	Horse	151	19.2	19.0	16.3	19.5	6.0k	21.6	20.7	19.2	21.7	22.6	22.3
	Lighthouse	309	16.6	16.5	13.6	17.5	8.3k	19.0	16.6	18.9	18.6	19.5	20.5
	Playground	307	19.1	19.2	11.4	18.7	7.7k	17.9	16.5	11.3	20.4	21.2	21.0
	Train	301	16.8	15.4	13.8	16.2	12.6k	19.6	14.4	15.6	18.5	19.8	18.5
	Average	254	18.8	18.3	15.6	18.5	7.8k	19.9	18.1	17.8	19.8	21.1	21.0
Avg Time			32min	2min	3min	1.3h		48h	11h	14min	2.2h	5h	1h
Advanced	Auditorium	302	19.6	12.2	16.7	18.7	13.6k	13.7	-	16.6	20.0	21.4	19.8
	Ballroom	324	16.3	18.3	13.1	17.9	10.8k	17.2	-	10.4	18.9	18.0	15.6
	Courtroom	301	18.2	17.2	12.3	17.1	12.6k	14.6	-	10.2	16.3	18.7	17.8
	Palace	509	14.2	11.7	10.8	10.7	21.9k	13.8	-	8.6	11.0	15.3	12.3
	Temple	302	18.1	15.7	11.8	9.7	17.5k	13.3	-	11.9	14.8	19.6	16.1
	Average	348	17.3	15.0	12.9	14.8	15.6k	14.5	-	11.5	16.2	18.6	16.3
Avg Time			1h	2min	4min	1h		71h		27min	2.8h	10h	2.1h

Initialize with a sparse COLMAP reconstruction from 150-500 images, and extend and refine it using all available frames.

Tanks and Temples from Light3R-SfM[3]

	Method	Align.	RRA@5 ↑	RTA@5 ↑	ATE ↓	Reg. ↑	Time [s] ↓
25	COLMAP	OPT	13.7	12.6	0.038	44.4	-
	GLOMAP	OPT	58.4	53.6	0.076	86.1	16.1
	ACE0	OPT	1.2	1.4	0.112	100.0	1042.7
	DF-SfM	OPT	47.5	48.7	0.081	99.4	-
	FlowMap	OPT	0.7	1.5	0.107	100.0	-
	VGGSfM	OPT	55.7	57.4	0.058	96.2	135.7
	MASt3R-SfM	OPT	68.0	70.3	0.034	100.0	283.2
	Spann3r	FFD	19.6	30.7	0.060	100.0	8.3
200	Light3R-SfM	FFD	50.9	54.2	0.048	100.0	4.4
	COLMAP	OPT	64.7	57.7	0.019	97.0	-
	GLOMAP	OPT	73.5	74.8	0.016	100.0	536.7
	ACE0	OPT	55.7	57.4	0.019	100.0	4604.4
	DF-SfM	OPT	66.8	69.3	0.016	33.3	-
	FlowMap	OPT	22.2	25.8	0.024	100.0	-
	VGGSfM	OPT	84.5	86.3	0.007	47.6	1511.6
	MASt3R-SfM	OPT	68.2	68.4	0.013	100.0	1609.0
full	Spann3r	FFD	22.8	28.6	0.019	100.0	60.4
	Light3R-SfM	FFD	52.4	53.1	0.016	100.0	33.4
	COLMAP	OPT	GT	GT	GT	GT	-
	GLOMAP	OPT	75.8	76.7	0.010	100.0	1977.7
	ACE0	OPT	56.9	57.9	0.015	100.0	5499.5
	DF-SfM	OPT	69.6	69.3	0.014	76.2	-
	FlowMap	OPT	31.7	35.7	0.017	66.7	-
	VGGSfM	OPT	-	-	-	0.0	2134.2
	MASt3R-SfM	OPT	49.2	54.0	0.011	100.0	2723.1
	Spann3r	FFD	20.3	24.7	0.016	100.0	116.2
	Light3R-SfM	FFD	52.0	52.8	0.011	100.0	63.4

Compare:

1. Relative pose accuracy RRA@5 and RTA@5
2. Absolute translation error (ATE)
3. Registration rate (Reg.)

* COLMAP, VGGSfM can reject hard images.

worst and best

ETH-3D from MAST3R-SfM[4]

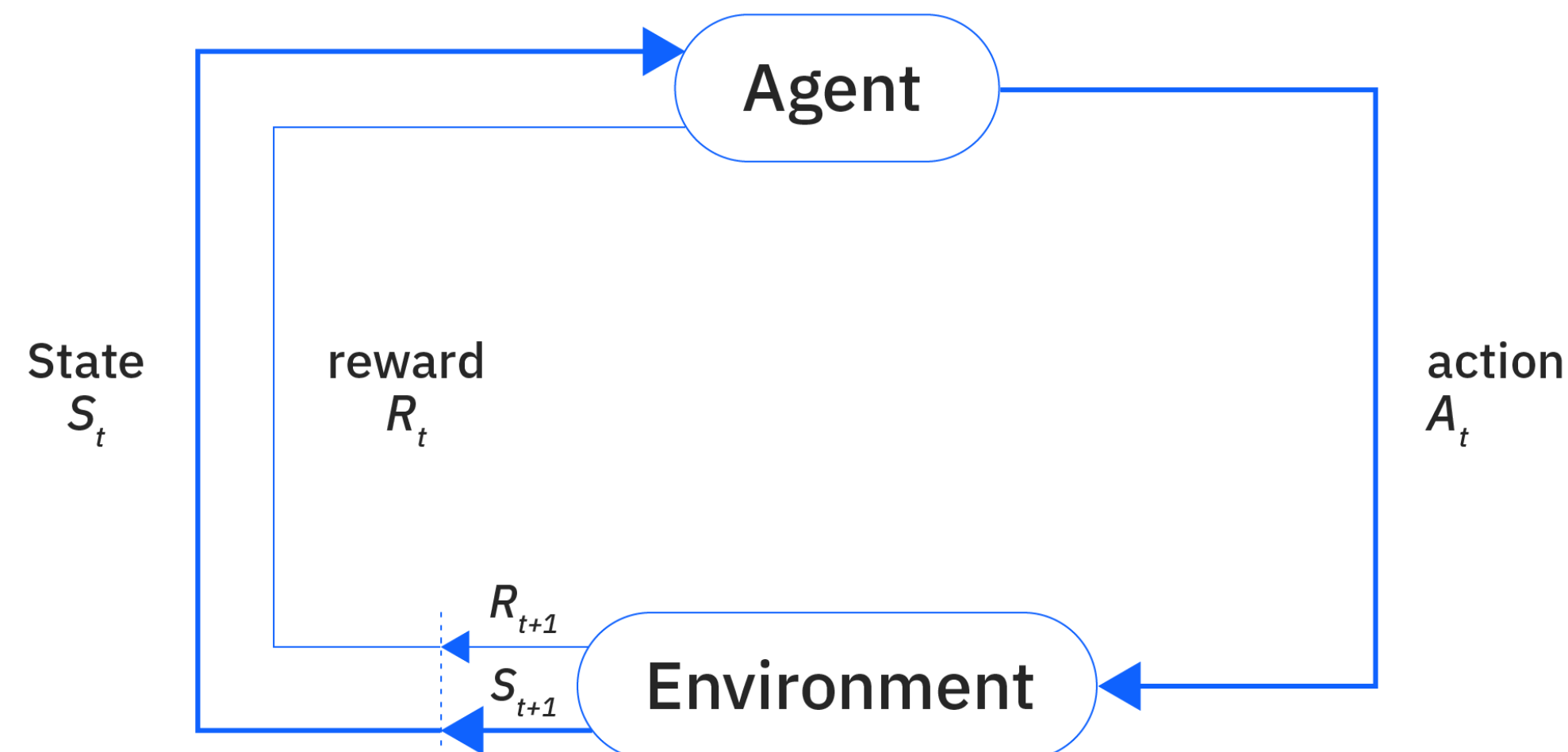
Scenes	COLMAP [46]		ACE-Zero [9]		FlowMap [50]		VGGSfM [62]		DF-SfM [20]		MASt3R-SfM	
	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5	RRA@5	RTA@5
courtyard	56.3	60.0	4.0	1.9	7.5	3.6	50.5	51.2	80.7	74.8	89.8	64.4
delivery area	34.0	28.1	27.4	1.9	29.4	23.8	22.0	19.6	82.5	82.0	83.1	81.8
electro	53.3	48.5	16.9	7.9	2.5	1.2	79.9	58.6	82.8	81.2	100.0	95.5
facade	92.2	90.0	74.5	64.1	15.7	16.8	57.5	48.7	80.9	82.6	74.3	75.3
kicker	87.3	86.2	26.2	16.8	1.5	1.5	100.0	97.8	93.5	91.0	100.0	100.0
meadow	0.9	0.9	3.8	0.9	3.8	2.9	100.0	96.2	56.2	58.1	58.1	58.1
office	36.9	32.3	0.9	0.0	0.9	1.5	64.9	42.1	71.1	54.5	100.0	98.5
pipes	30.8	28.6	9.9	1.1	6.6	12.1	100.0	97.8	72.5	61.5	100.0	100.0
playground	17.2	18.1	3.8	2.6	2.6	2.8	37.3	40.8	70.5	70.1	100.0	93.6
relief	16.8	16.8	16.8	17.0	6.9	7.7	59.6	57.9	32.9	32.9	34.2	40.2
relief 2	11.8	11.8	7.3	5.6	8.4	2.8	69.9	70.3	40.9	39.1	57.4	76.1
terrace	100.0	100.0	5.5	2.0	33.2	24.1	38.7	29.6	100.0	99.6	100.0	100.0
terrains	100.0	99.5	15.8	4.5	12.3	13.8	70.4	54.9	100.0	91.9	58.2	52.5
Average	49.0	47.8	16.4	9.7	10.1	8.8	65.4	58.9	74.2	70.7	81.2	79.7

worst and best

- Neither ACE0 nor FlowMap can handle **non-video** setups.
- COLMAP and VGGSfM are high sensitivity to not having highly overlapping images
=> meaning that these methods cannot really handle truly unconstrained collections (?)

Appendix - IQA with meta-reinforcement learning

- Re-define Image Quality Assessment(IQA) as:
A measure of how images are amenable **with respect to a given downstream task**, or task amenability.
=> task-specific, adaptable IQA
- Bi-level optimization of:
 - 1.) Task Predictor: Perform downstream task such as segmentation, classification.
 - 2.) IQA Controller: Select or weight images based on the prediction on task amenability.
- Reinforcement Learning:



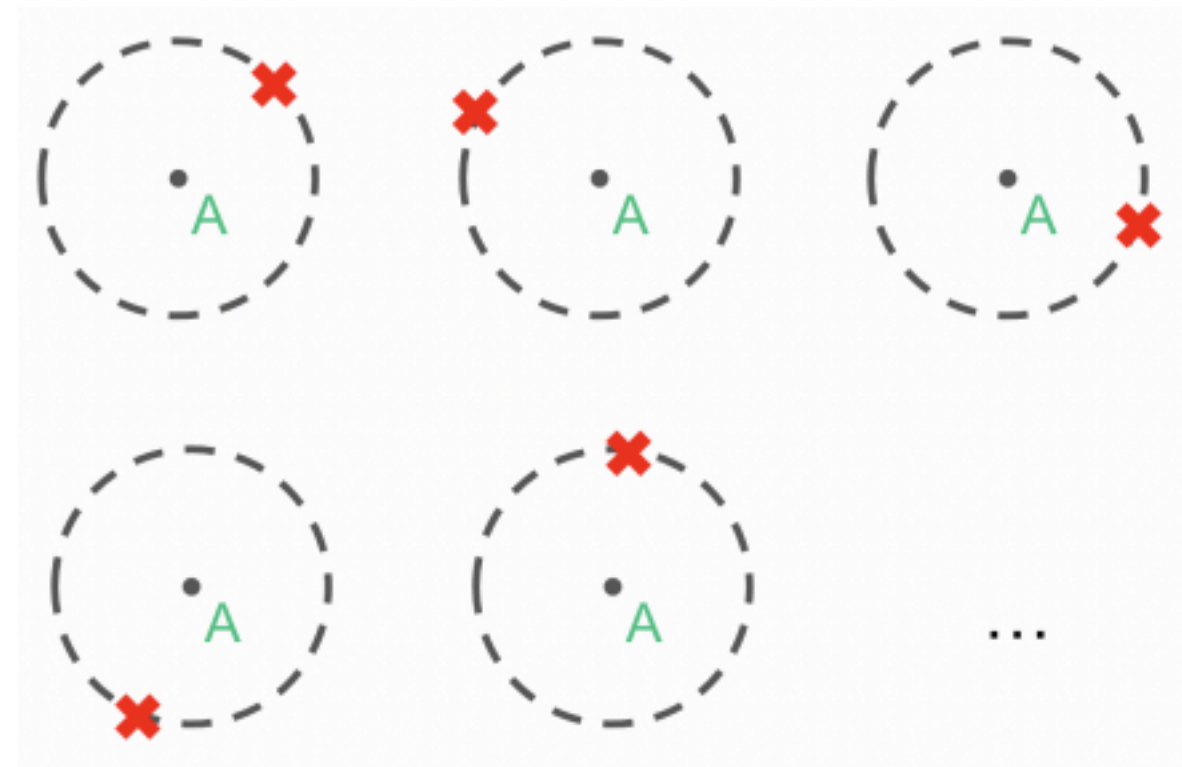
The future is independent of the past given the present.

A state S_t is Markov iff:

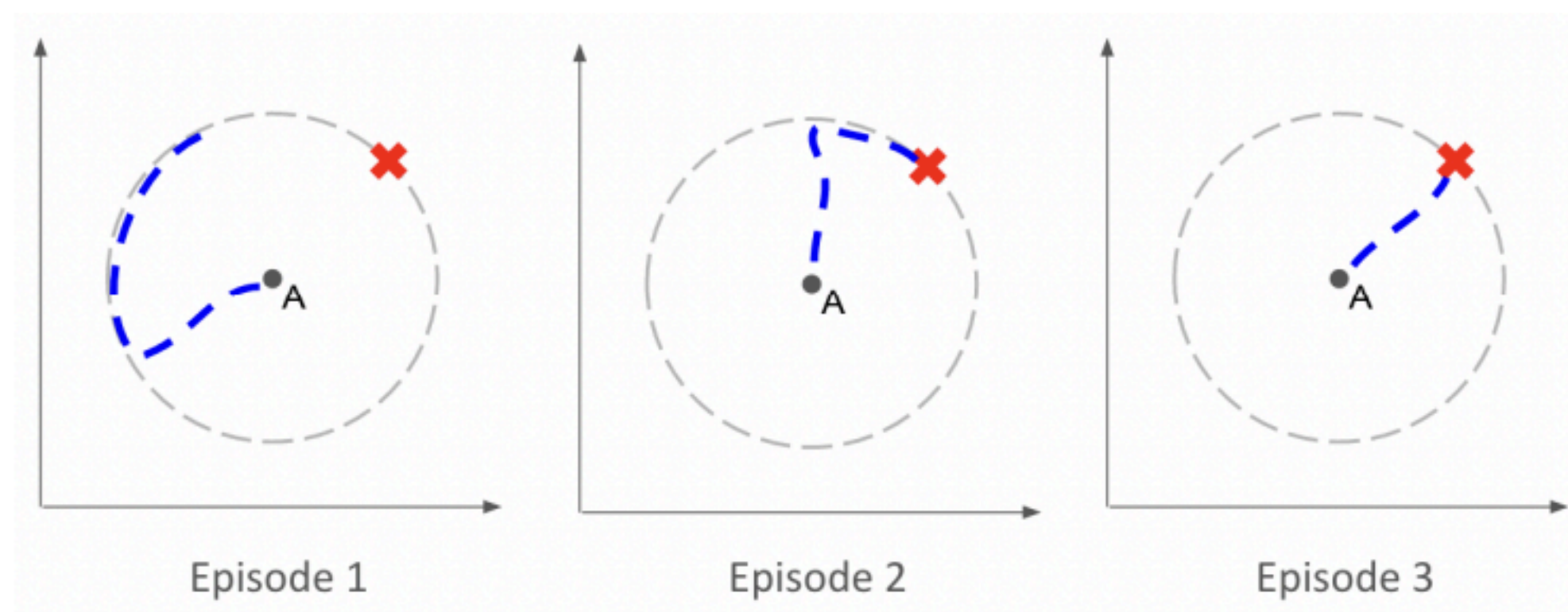
$$\mathbb{P}[S_{t+1} | S_t] = \mathbb{P}[S_{t+1} | S_1, \dots, S_t]$$

Appendix - IQA with meta-reinforcement learning

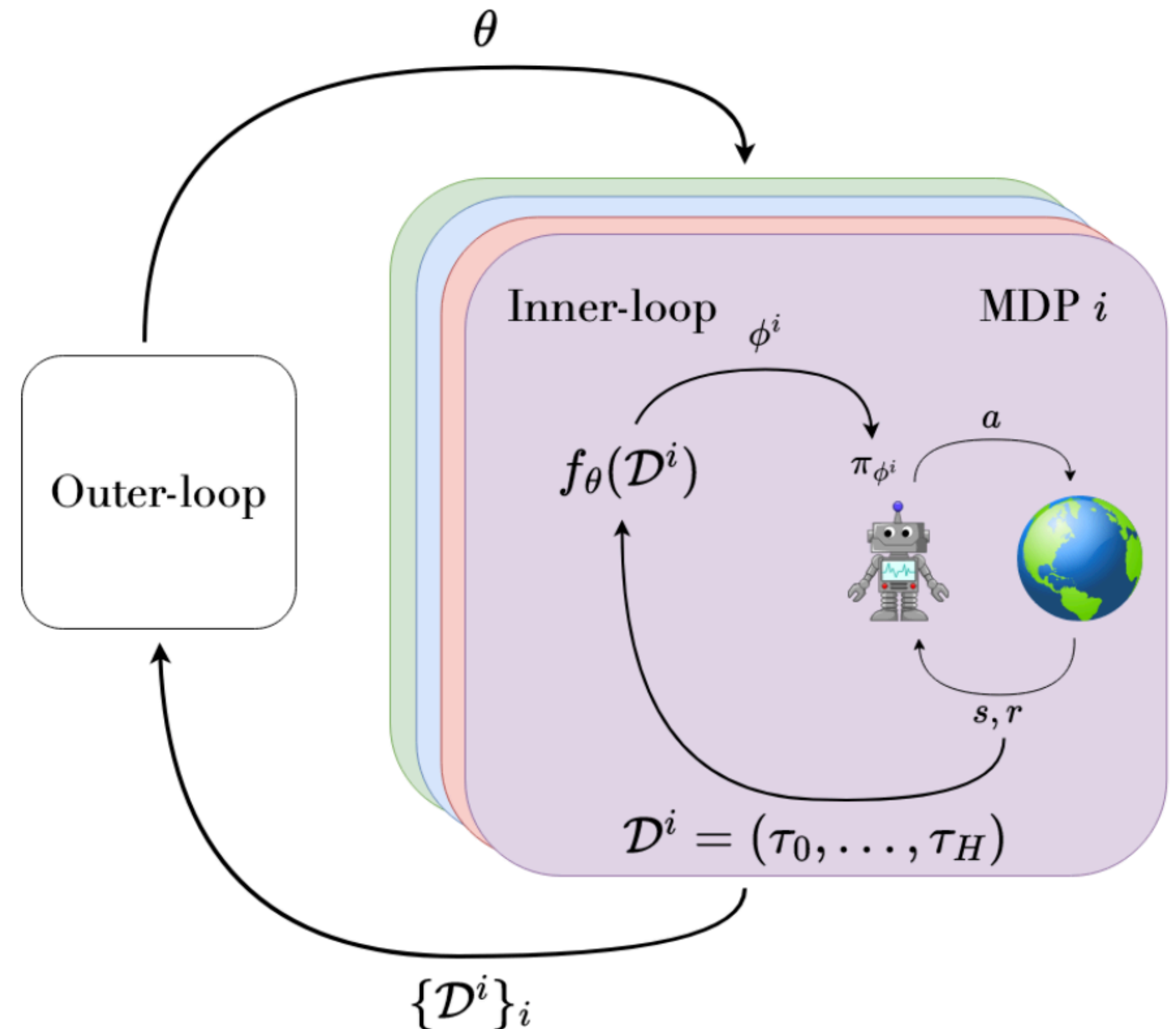
○ Meta-Reinforcement Learning



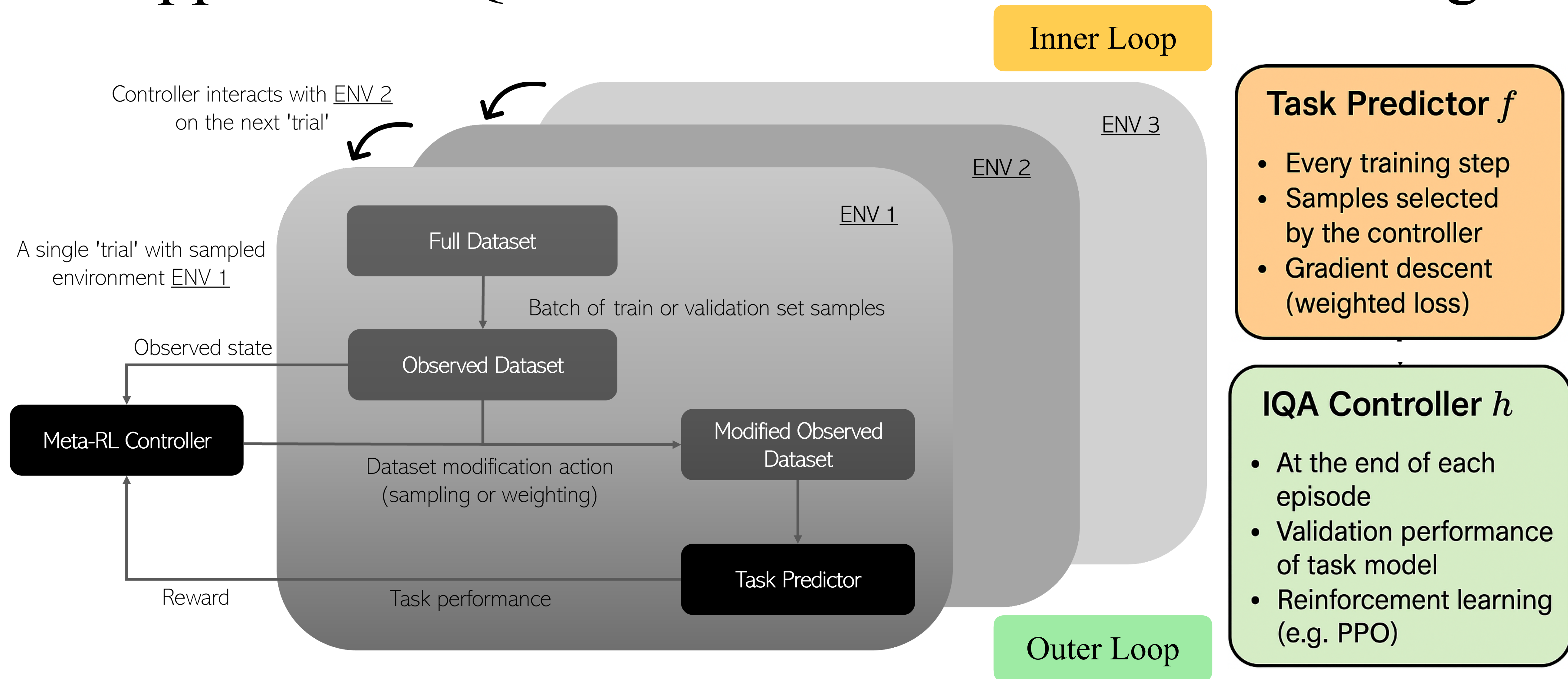
(a) Meta-Training Tasks



(b) Rollout at Meta-Test Time



Appendix - IQA with meta-reinforcement learning



Appendix - IQA with meta-reinforcement learning

Task	Reward computation strategy	Mean $\pm$ St.D.
Prostate presence (Acc.)	Non-selective baseline	0.897 ± 0.010
	$\tilde{R}_{\text{avg},t}$, fixed validation set	0.935 ± 0.014
	$\tilde{R}_{\text{w},t}$, weighted validation set	0.926 ± 0.012
	$\tilde{R}_{\text{sel},t}$, selective validation set	0.913 ± 0.012
Prostate segmentation (Dice)	Non-selective baseline	0.815 ± 0.018
	$\tilde{R}_{\text{avg},t}$, fixed validation set	0.890 ± 0.017
	$\tilde{R}_{\text{w},t}$, weighted validation set	0.893 ± 0.018
	$\tilde{R}_{\text{sel},t}$, selective validation set	0.865 ± 0.014
Pneumonia detection (Acc.)	Non-selective baseline	0.817 ± 0.026
	$\tilde{R}_{\text{w},t}$, weighted validation set (DDPG)	0.843 ± 0.033
	$\tilde{R}_{\text{w},t}$, weighted validation set (PPO)	0.838 ± 0.031

References

- [1] Brachmann, E., Wynn, J., Chen, S., Cavallari, T., Monzpart, '., Turmukhambetov, D., & Prisacariu, V. (2024). Scene Coordinate Reconstruction: Posing of Image Collections via Incremental Learning of a Relocalizer. *European Conference on Computer Vision*.
- [2] Saeed, S. U., Fu, Y., Stavrinides, V., Baum, Z. M. C., Yang, Q., Rusu, M., Fan, R. E., Sonn, G. A., Noble, J. A., Barratt, D. C., & Hu, Y. (2022). Image quality assessment for machine learning tasks using meta-reinforcement learning. *Medical image analysis*, 78, 102427. <https://doi.org/10.1016/j.media.2022.102427>
- [3] Sven Elflein, Qunjie Zhou, Sérgio Agostinho, & Laura Leal-Taixé. (2025). Light3R-SfM: Towards Feed-forward Structure-from-Motion.
- [4] Bardienus Duisterhof, Lojze Zust, Philippe Weinzaepfel, Vincent Leroy, Yohann Cabon, & Jerome Revaud. (2024). MAST3R-SfM: a Fully-Integrated Solution for Unconstrained Structure-from-Motion.